

Downburst wind loading on bridge decks through an analytical model

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ABSTRACT

A mathematical model is proposed for estimating the wind loads caused by the passage of a downburst through horizontal structures, such as bridge decks. The model simulates the three-dimensional flow fields of stationary and environmentally-driven downbursts. It is based on radial and vertical shaping functions of the velocity components, following a path developed for aircraft structures. The radial shaping function of the horizontal velocity is defined with attention to reproducing the local increase in velocity observed in the average field measurements. In addition, the coefficients of an existing empirical vertical shaping function of the horizontal velocity are re-calibrated to better fit the available field data. The shaping functions of the vertical components are derived by imposing mass conservation.

The model allows for predicting the large angles of attack on the decks during the passage of downbursts with small to medium diameters. The occurrence of such large angles of attack also implies a change in the aerodynamic design procedure of bridge decks, for which wind tunnel tests are commonly conducted only for angles of attack in the range of about $\pm 10^\circ$. The aerodynamic coefficients of a NACA profile and a pseudo-rectangular section are used due to the absence in the literature of aerodynamic coefficients of bridge decks for very large angles of attack. Several runs are conducted by varying the downburst properties and the deck height above the ground to provide results useful for the aerodynamic design of bridges. The results confirm the key role played by the vertical component of the velocity in developing the wind loads on the decks during the passage of a downburst. Non-negligible wind loads are obtained, which are not predictable with the existing approaches. The need for conducting dynamic analyses of flexible bridges is also emerging.

1. Introduction

International codes and standards for wind loading on structures are based on traditional atmospheric boundary layer (sheared) flows. These are generated by extra-tropical cyclones of synoptic scale, which can even reach 1000 km of horizontal dimension and take some days for their passage. Thus wind direction is constant for some hours and the flow velocity is almost horizontal increasing from the ground in a logarithmic way. Turbulence is mainly generated by the friction at the ground level. Atmospheric boundary layer flows are modelled as stationary, ergodic, Gaussian processes with significant simplifications in their mathematical treatment [1].

Several shreds of evidence in many parts of the world have shown that extreme events are produced by thunderstorms [2,3]. They are convective phenomena that produce small disturbances compared to extra-tropical cyclones but can generate strong winds through downbursts and tornadoes [4]. Limiting the discussion to downbursts, at the end of their life cycle (see [5] for more details), strong downdrafts that reach the ground (like a jet) are produced generating strong winds for

a short time (5–10 min). According to Fujita [6], downbursts can be further subdivided into macrobursts and microbursts, referring to the latter when the horizontal dimension is below 4 km.

Microbursts have been first studied in the aeronautical field to prevent aircraft downfalls during the landing and take-off phases. A microburst is considered the cause of the famous aircraft accident that occurred on the 24th of June 1975 at the JFK airport in New York, in which 113 people died [5]. Unfortunately, for about two decades, up to the development of modern alarm systems, microbursts have been the cause of many other aircraft accidents worldwide.

Downbursts show remarkably three-dimensional velocity fields, in which the environmental flow is superimposed to the jet flow. This means that by recording the wind direction and velocity at a point, significant variations are observed both in wind speed and yaw and pitch angles. The maximum velocity profile is located at a certain distance from the downburst centre (about 1 jet diameter) and presents a maximum at 30–100 m over the ground according to the first field

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measurements [7,8], which can be extended to 10–200 m by considering the recent experimental tests [9–11]. Hence, larger forces on low and mid-rise buildings are expected than in the case of boundary layer winds. In addition, downbursts are non-stationary processes in which the turbulent fluctuations are added to a running mean. As a consequence of the transient nature of the phenomenon, fast and simple analyses in the frequency domain are not possible leaving the time domain the most straightforward analysis option, though more refined evolutionary spectral models have been applied to the estimate of response to thunderstorms (e.g. [12]). Experimental results have shown that surface pressure fluctuations on mid-rise buildings seem to be more correlated than those of the boundary layer winds [13].

For the reasons above, codes and standards should also account for thunderstorm downburst phenomena in defining the wind loads on structures. Nevertheless, just in recent years, microburst effects on civil structures captured the attention of the wind engineering community.

Research efforts have been concentrated on the estimation of the wind loads and response for vertical structures, such as single-degree-of-freedom systems [14], low-rise buildings (cubes and portal frames) [15], lattice transmission towers [16], uniformly cantilevered towers [17], telecommunication antenna masts and towers [18], steel lighting poles [19], high-rise buildings [20,21]. Large responses of long-span bridges have been observed during non-synoptic events, possibly exceeding the ones observed from stationary wind [22]. To the Authors' knowledge, only one work, approaching the problem by CFD, deals with thunderstorm downburst effects on long-span bridges [23]. Since downburst parameters are quite uncertain, it is recommended that several cases are preliminarily studied to select the worst scenario for the design. The use of simplified analytical models is then a valuable tool for parametric analyses.

The simulation of the vertical component of the wind velocity in case of downbursts is not required for estimating wind loads and dynamic responses of vertically developed structures. Thus, researchers have been mainly focused on the development of models for the maximum wind velocity profiles by considering the horizontal velocity components only [9,24,25]. Conversely, for bridge decks, the vertical component combined with the horizontal one results in significant changes in the angle of attack, which plays a key role in the estimation of both wind loads and dynamic response. The issue of the critical role of the angle of attack in bridge response to downburst has been raised in a previous paper [26] where some specific considerations on the collapse of Polcevera Bridge in Genoa were reported. Three-dimensional models able to simulate also the vertical components of the wind velocity field are necessary for estimating the response of horizontal structures, such as bridge decks and roofs. This necessity implies also a change in the aerodynamic design procedure of bridge decks, for which wind tunnel tests are commonly conducted for angles of attack in the range ± 10 degrees.

The question is whether a combination of angle of attack and flow velocity exists that may result in aerodynamic forces that may endanger a bridge. It is indeed one of the purposes of the paper to highlight the potential issue related to the angle of attack, especially for streamlined cross-sections that are optimized for small angles of attack.

Three-dimensional models have been developed in the aeronautical field to simulate approaches to landing and escape manoeuvres for aircraft to survive inadvertent downburst encounters. The first analytical model for non-turbulent downburst wind speeds was an axis-symmetric, steady-state model based on shaping functions to satisfy the mass continuity equation and simulate boundary layer effects [27]. The model considerably overestimates the horizontal winds in the radial direction and underestimates the vertical one compared with experimental and numerical results. These drawbacks have been partially solved by changing the shaping functions [28,29] but non-negligible differences with the available data [30–32] are still present. The model has been extended to consider also the boundary layer effect [33]. It was also changed by considering Wood's expression [9] for the

vertical variation of the horizontal velocity [34] but without a continuity equation to regulate the two wind velocity components. A more refined version has been developed by using Holmes's expression for the radial variation of the horizontal velocity [24] and Wood's for the vertical profile [9]. The numerical integration required for obtaining the shaping functions weighting down the practical implementation of the model [35]; the profile of the horizontal wind speed accounts for the boundary layer growth without satisfying the mass conservation equation. The model also presents a lack in the prediction of the profile of the vertical velocity component. A model that tackles the previous issues has been developed employing simple shape functions able to fit the field measures respecting the mass continuity equation and including the nonlinear effects caused by the atmospheric boundary layer growth [36].

The results from full-scale measurements available in the literature [7], although limited in number, show a local increase in the radial variation of the horizontal velocity caused by the presence of the radial flow, as also highlighted in the schematic drawing shown in Fig. 1, for r/r_{max} in the range from 1.5 to 2.0. This feature has not been simulated in the existing models.

In this work, a new three-dimensional model is formulated by following the path traced by Oseguera and Bowles [27], which is based on separated-variable radial and vertical modulations of the wind velocity components. The vertical and radial variations of the horizontal velocity component are defined in the new model as the best fit of the full-scale data including the local increase in velocity in the radial variation. The vertical and radial shape functions of the vertical wind velocities have been derived according to the mass conservation equation.

The passage of the downburst produces a wind velocity field involving very large variations of the angles of attack at the common bridge deck heights. The model has been applied to both the NACA 0015 wing and a pseudo-rectangular 3:1 cross-sections. The choice of such sections is due to the lack of aerodynamic coefficient measurements for bridge decks at very large angles of attack in the literature. Although unrealistic for a bridge deck these cross-sections represent extreme behaviours, one very streamlined with low drag and large lift variations, the other much bluffer with high drag.

In the next section, the wind velocity field model is outlined, in which two subsections are dedicated to the environmental and downburst velocity fields. The analytical formulations for the stationary downburst are reported in Section 3 where the relationships of the new shaping functions are derived. Section 4 refers to the numerical simulations of the passage of a downburst through a bridge deck carried out by considering the proposed model and Oseguera-Bowles-Vicroy model [27–29]. Finally, some concluding remarks are reported at the end of the paper.

2. Wind velocity field

We consider a Cartesian reference frame ($\mathbf{O}; x, y, z$), with unit vectors $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$, where the origin $\mathbf{O}$ is set at ground level at one end of the bridge main span, the y -axis oriented along the bridge span, and the z -axis vertical and pointing upwards. The wind velocity field $\mathbf{V}(x, y, z; t)$, variable with time t , can be written as the sum of three terms: (1) the *environmental* velocity field $\mathbf{U}(z)$, (2) the *downburst* velocity field $\mathbf{V}(x, y, z; t)$, and (3) the fluctuating *turbulent* contribution $\mathbf{v}(x, y, z; t)$; namely

$$\mathbf{V}(x, y, z; t) = \mathbf{U}(z) + \mathbf{V}(x, y, z; t) + \mathbf{v}(x, y, z; t),$$

2.1. Environmental velocity field

The environmental velocity is horizontal and directed at an angle θ with respect to the x -axis, it represents the mean wind and is also responsible for the 'transport' of the downburst. Let C_0 be the projection

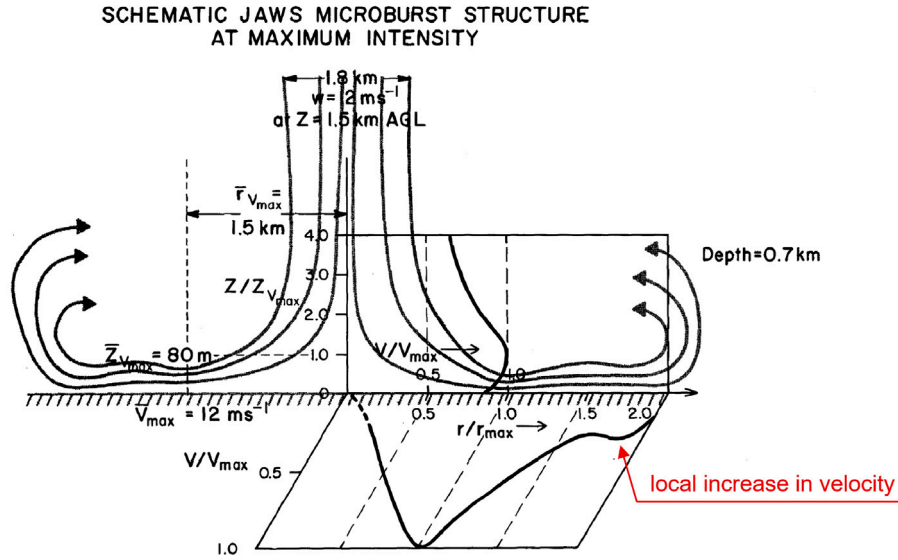


Fig. 1. Schematic microburst structure with highlighted local increase of velocity ("bulge") [7].

of the downburst centre on the ground at time $t = 0$ and $\mathbf{e}_1 = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$ the unit vector in the direction of the environmental wind. The environmental velocity, is, in the general case, variable with the height from the ground z , namely

$$\mathbf{U} = U(z)\mathbf{e}_1.$$

Here, we further assume that the environmental velocity is constant with height and equal to the velocity of the moving downburst U_r , i.e. $U(z) = U_r$. This simplifying assumption is supported by the relatively modest values of environmental wind velocities, which are lower than characteristic values of synoptic wind and thus prevent the development of an atmospheric boundary layer.

2.2. Downburst velocity field

The coordinates of the centre of the downburst at time t are

$$\begin{cases} x_C(t) = x_{C0} + U_r t \cos \theta, \\ y_C(t) = y_{C0} + U_r t \sin \theta, \end{cases}$$

where x_{C0} and y_{C0} are the coordinates of the centre of the downburst at time $t = 0$. The stationary downburst velocity field is given by

$$\mathbf{V}(x, y, z; t) = \mathcal{V}_r(r(x, y; t), z)\mathbf{n}_r(x, y; t) + \mathcal{V}_z(r(x, y; t), z)\mathbf{k},$$

where $\mathcal{V}_r$ and $\mathcal{V}_z$ are the radial and vertical downburst velocity components,

$$r(x, y; t) = \sqrt{(x - x_C(t))^2 + (y - y_C(t))^2}$$

is the distance from the downburst centre, and

$$\mathbf{n}_r(x, y, t) = \frac{x - x_C(t)}{r(x, y; t)}\mathbf{i} + \frac{y - y_C(t)}{r(x, y; t)}\mathbf{j},$$

the radial unit vector.

The radial and vertical velocities are written as

$$\mathcal{V}_r(r, z) = U_{\max} v_\rho(r/r_{\max}) v_\zeta(z/z_{\max}), \quad (1)$$

$$\mathcal{V}_z(r, z) = -W_{\max} \omega_\rho(r/r_{\max}) \omega_\zeta(z/z_{\max}), \quad (2)$$

where v_ρ and v_ζ are modulating functions of the radial velocity as a function of the dimensionless radial and vertical coordinates $\rho = r/r_{\max}$ and $\zeta = z/z_{\max}$, ω_ρ and ω_ζ are modulating functions of the vertical velocity, $r_{\max}$ and $z_{\max}$ are the radial and vertical positions at which the maximum downburst horizontal velocity $U_{\max}$ is achieved, $W_{\max}$ is the maximum downburst vertical downward velocity; the latter is achieved

at the downburst centre and can be obtained as $W_{\max} = c \frac{z_{\max}}{r_{\max}} U_{\max}$, where the coefficient c depends on the downburst model (see later, Eqs. (11a) and (A.1)).

2.3. Turbulent velocity field

The turbulent contribution has a general form

$$\mathbf{v} = u(x, y, z; t)\mathbf{e}_1 + v(x, y, z; t)\mathbf{e}_2 + w(x, y, z; t)\mathbf{k}.$$

where u , v and w are the multicorrelated zero-mean turbulent velocity components and $\mathbf{e}_2 = \mathbf{k} \times \mathbf{e}_1$. The turbulence effect is not considered in the present paper.

2.4. Downburst parameters

To summarize the downburst parameters are:

- $U_{\max}$ downburst maximum horizontal velocity;
- $r_{\max}$ distance from centre of downburst maximum velocity;
- $z_{\max}$ height of maximum horizontal velocity of the stationary downburst;
- U_r environmental velocity;
- θ direction of downburst motion w/r to bridge;
- (x_{C0}, y_{C0}) position of downburst centre at time $t = 0$;
- A model for the turbulent contribution (not used here).

3. Analytical formulation for stationary downbursts

A stationary downburst is considered. The flow is assumed as incompressible. The mass conservation equation reads

$$\text{div } \mathbf{V} = 0, \quad (3)$$

being $\mathbf{V}$ is the velocity vector. The flow is assumed as axial symmetric and Eq. (3) is written in polar coordinates as

$$\frac{1}{r} \frac{\partial(\mathcal{V}_r r)}{\partial r} + \frac{\partial \mathcal{V}_z}{\partial z} = 0. \quad (4)$$

We assume that the downburst is characterized by the maximum horizontal velocity $U_{\max}$ and by the height from the ground $z_{\max}$ and distance from the downburst centre $r_{\max}$ at which it occurs. We introduce the dimensionless coordinates

$$\rho = \frac{r}{r_{\max}}, \quad (5a)$$

$$\zeta = \frac{z}{z_{\max}}, \quad (5b)$$

so that Eq. (4) is rewritten as

$$\frac{1}{\rho} \frac{\partial(\mathcal{V}_\rho \rho)}{\partial \rho} + \frac{r_{\max}}{z_{\max}} \frac{\partial \mathcal{V}_\zeta}{\partial \zeta} = 0, \quad (6)$$

where $\mathcal{V}_\rho$ and $\mathcal{V}_\zeta$ are the velocities components seen as functions of the dimensionless coordinates.

Similarly to [27], we consider solutions with separate variables in the form

$$\mathcal{V}_\rho(\rho, \zeta) = A \frac{f(\rho^2)}{\rho} \frac{dg(\zeta)}{d\zeta}, \quad (7a)$$

$$\mathcal{V}_\zeta(\rho, \zeta) = -2A \frac{z_{\max}}{r_{\max}} \frac{df(\rho^2)}{d(\rho^2)} g(\zeta), \quad (7b)$$

where A is a constant, and $f(\rho^2)$ and $g(\zeta)$ are differentiable functions.

To simplify the comparison with empirical formulations of the horizontal velocity component, we rewrite Eq. (7a) as

$$\mathcal{V}_\rho(\rho, \zeta) = \mathcal{U}_{\max} v_\rho(\rho) v_\zeta(\zeta), \quad (8)$$

where v_ρ and v_ζ are modulating functions which have a unit maximum values at $\rho = 1$ and $\zeta = 1$ respectively. By comparing Eqs (8) and (7a), we have

$$A = \mathcal{U}_{\max}, \quad (9a)$$

$$v_\rho(\rho) = \frac{f(\rho^2)}{\rho}, \quad (9b)$$

$$v_\zeta(\zeta) = \frac{dg(\zeta)}{d\zeta}. \quad (9c)$$

We aim for an expression for the vertical component of the downburst velocity similar to the one of Eq. (8), namely

$$\mathcal{V}_\zeta(\rho, \zeta) = -\mathcal{W}_{\max} \omega_\rho(\rho) \omega_\zeta(\zeta), \quad (10)$$

where $\mathcal{W}_{\max}$ is the maximum vertical velocity (expected in the centre of the downburst at large heights from the ground), and ω_ρ and ω_ζ are modulating functions.

By comparing Eqs (10) and (7b), we have

$$\mathcal{W}_{\max} = 2\phi_0 g_{\max} \frac{z_{\max}}{r_{\max}} A, \quad (11a)$$

$$\omega_\rho(\rho) = \frac{1}{\phi_0} \frac{df(\rho^2)}{d(\rho^2)}, \quad (11b)$$

$$\omega_\zeta(\zeta) = \frac{g(\zeta)}{g_{\max}}, \quad (11c)$$

where $\phi_0 = \frac{df}{d(\rho^2)}(0)$ and $g_{\max} = \max_{\zeta>0} (g(\zeta))$.

In the following sections, we choose heuristic analytical forms for the functions f and g and calibrate them on field measures.

3.1. Radial modulation of the horizontal velocity

As to the horizontal modulation, we assume a parametric analytical expression for f , able to reproduce the measured shape heuristically and requiring the boundary condition $f(0) = 0$, namely $f(\rho^2) = a_1 \rho^2 (e^{-a_2 \rho^{2a_3}} + a_4 e^{a_5 - a_6 \rho^{2a_7}})$, resulting in

$$v_\rho(\rho) = a_1 \rho (e^{-a_2 \rho^{2a_3}} + a_4 e^{a_5 - a_6 \rho^{2a_7}}), \quad (12)$$

where a_i , $i = 1, 2, \dots, 7$ are parameters. By imposing the maximum of $v_\rho(\rho)$ equal to unity at $\rho = 1$, namely

$$\begin{cases} \frac{dv_\rho}{d\rho}(1) = 0, \\ v_\rho(1) = 1, \end{cases}$$

one gets

$$\begin{cases} a_1 = \frac{1}{e^{-a_2} + a_4 e^{(a_5 - a_6)}}, \\ a_7 = \frac{e^{a_6 - a_2 - a_5} (1 - 2a_2 a_3) + a_4}{2a_4 a_6}, \end{cases} \quad (13)$$

the parameters are estimated by least square optimization on the field-measurement data by [7], obtaining $a_2 \approx 0.1624$, $a_3 \approx 4.0439$, $a_4 \approx 0.2436$, $a_5 \approx 0.1340$, and $a_6 \approx 0.00481$. From Eq. (13) one gets also $a_1 \approx 0.8871$ and $a_7 \approx 4.0545$. The shaping function is depicted in Fig. 2 together with the field recordings used for the fitting.

It is evident the good agreement of the fitted shaping function (red solid curve) with the field-measurement average (solid black curve). It is able to follow the aforementioned local increase in velocity recorded in the field measurements. In addition, the shaping functions of other existing models are plotted in the same figure for comparisons. In particular, the shaping function of the first three-dimensional model [27] shows a significant overestimation of the field-measurement average, which is corrected in its modified version [29]. The model developed by Holmes [24] presents the best fitting ahead of the maximum horizontal velocity ($\rho = 1$) but a non-negligible underestimation of the field-measurement average in the descending part. Vicroy's model presents also a good fitting (with the difference from ours for not reproducing the "bulge"), it is also differentiable and will be used for the comparisons.

3.2. Radial modulation of the vertical velocity

The radial modulation of the vertical velocity follows from Eq. (11b),

$$\omega_\rho = a_1 \left((1 - a_2 a_3 \rho^{2a_3}) e^{-a_2 \rho^{2a_3}} + a_4 (1 - a_6 a_7 \rho^{2a_7}) e^{a_5 - a_6 \rho^{2a_7}} \right),$$

where the value of $\phi_0 = a_1 (1 + a_4 e^{a_5}) \approx 1.1341$ has been used. The shaping function is represented in Fig. 2 with those derived in the literature [27,29] (Holmes model [24] does not consider the vertical velocity component). The Oseguera-Bowles model tends to zero as the radial coordinate increases, with a monotonically decreasing behaviour, while Vicroy's and the present model approach zero exhibiting one and three changes in slope, respectively.

3.3. Vertical modulation of the horizontal velocity

An empirical expression of the vertical profile of the horizontal velocity that fits field measures very well is the one proposed by [9], see also [19]. The function $u_\zeta(\zeta)$ is in the form

$$u_\zeta(\zeta) = b_1 \zeta^{b_2} \operatorname{erfc}(b_3 \zeta), \quad (14)$$

where $\operatorname{erfc}(x) := 1 - \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$ is the complementary error function and b_i , with $i = 1, 2, 3$, are parameters. By imposing a unit maximum for $\zeta = 1$, one gets

$$\sqrt{\pi} b_2 e^{b_3^2} \operatorname{erfc}(b_3) - 2b_3 = 0, \quad (15a)$$

$$b_1 = \frac{1}{\operatorname{erfc}(b_3)}, \quad (15b)$$

The parameter $b_2 = 1/6$ originally proposed by [9] is here reoptimized on field measures, getting $b_2 \approx 0.1981$; $b_1 \approx 1.2011$ and $b_3 \approx 0.1495$ follow from Eqs (15). The resulting modulating function is plotted in Fig. 3. A good agreement with the field-measurement average curve is apparent.

3.4. Vertical modulation of the vertical velocity

By integrating Eq. (9c), we get

$$g(\zeta) = b_1 \frac{\zeta^{1+b_2}}{1+b_2} \left(\operatorname{erfc}(b_3 \zeta) - \frac{b_3 \zeta}{\sqrt{\pi}} E_{-b_2/2}((b_3 \zeta)^2) \right) + b_4, \quad (16)$$

where $E_a(x) := \int_x^\infty \frac{e^{-xt}}{t^a} dt$ is the exponential integral function of parameter a , and b_4 is an integration constant that is determined by imposing a vanishing vertical velocity component at ground level (from which derives that $\lim_{\zeta \rightarrow 0^+} g(\zeta) = 0$); one has $b_4 \approx 5.2467$. The function

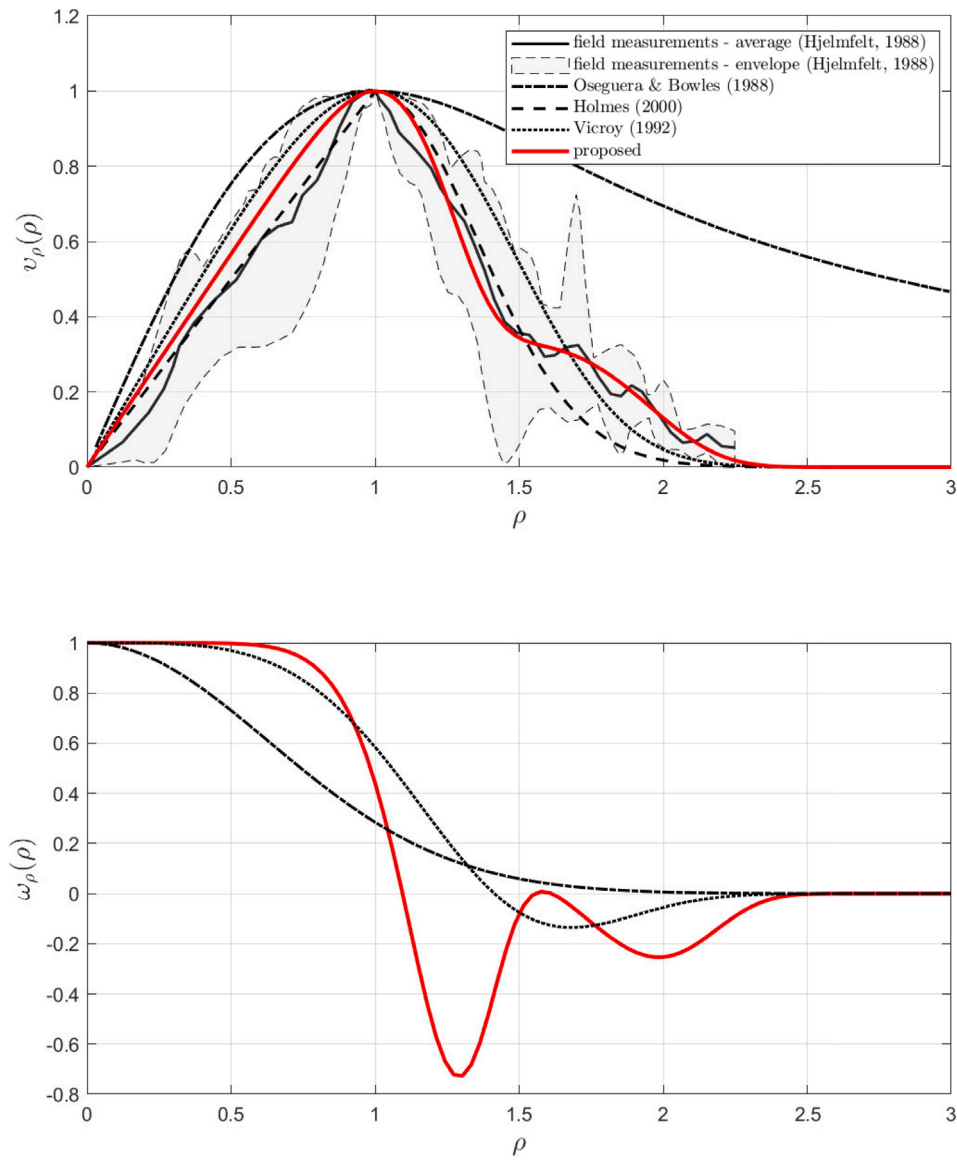


Fig. 2. Radial shaping functions: horizontal modulation as obtained by data fit (top); vertical modulation as derived from mass continuity equation (bottom); the shaping functions of three existing models [24,27,29] are also reported.

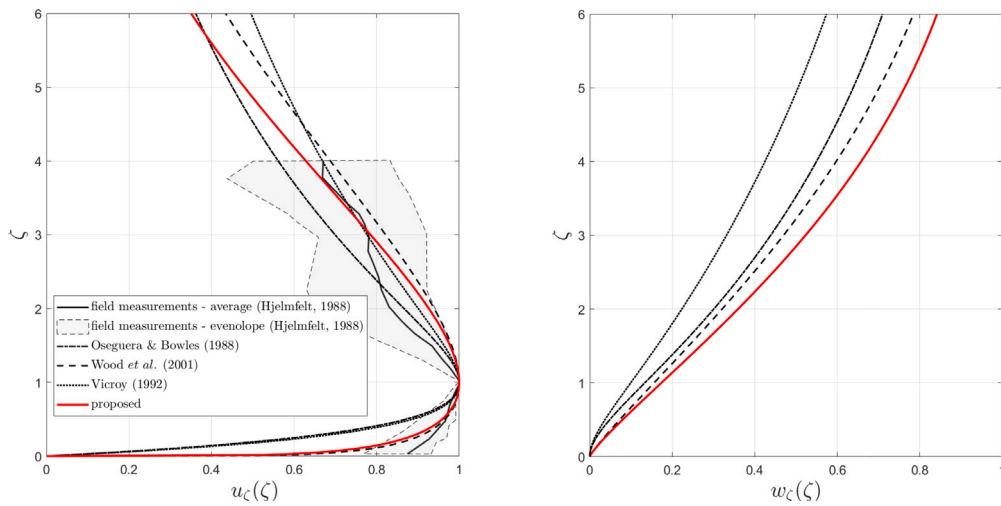


Fig. 3. Vertical shaping functions: horizontal velocity derived from data fit (left); b) vertical velocity derived from mass continuity equation (right); the shaping functions of three existing models [9,27,29] are also reported.

g is monotonically increasing and has an asymptotic value $g_{\max} = b_4$. Therefore, the vertical modulation of the vertical velocity can be obtained from Eq. (9c),

$$w_{\zeta}(\zeta) = 1 - \frac{b_1}{b_4} \frac{\zeta^{1+b_2}}{1+b_2} \left(\frac{b_3 \zeta}{\sqrt{\pi}} E_{-b_2/2}((b_3 \zeta)^2) - \operatorname{erfc}(b_3 \zeta) \right), \quad (17)$$

and it is plotted in Fig. 3. As for the radial variation of the vertical component, no field measurement is available for comparison. The present model provides the highest values of the vertical velocity modulation along the height and it is very close to the shaping function derived by imposing the satisfaction of the mass conservation function by Wood's profile [9], here computed for comparison.

As to the downdraft velocity, after calibration one has that

$$\mathcal{W}_{\max} = 11.90 \frac{z_{\max}}{r_{\max}} \mathcal{U}_{\max} \quad (18)$$

4. Simulations

In the present section, some simulation results are presented in terms of flow pattern, flow field and wind loads. The parameters of the simulated downbursts are obtained from literature data by considering the following assumptions:

- the peak velocity of a stationary downburst can be in the order of $\mathcal{U}_{\max} = 40$ m/s (roughly the value of 38.5 m/s reported by [19]);
- the environmental velocity U_t has been considered equal to one-third of the peak velocity according to [37];
- the height of the maximum radial velocity $z_{\max}$ ranges between 25 m and 120 m (roughly the interval 25–100 m reported by [19]);
- the distance from the centre of the downburst of the peak horizontal velocity for a stationary downburst $r_{\max}$ ranges between 400 m and 5500 m. These values have been obtained starting from the minimum/maximum values of $z_{\max}$ taking into account that $z_{\max} = (0.01-0.05) \cdot D$ [11,30,36,38] and $r_{\max} = (0.75-1.00) \cdot D$ [7].

Two extreme downburst configurations can be defined by considering the extremes of the aforementioned intervals:

- a small environmentally-driven downburst, characterized by $z_{\max} = 25$ m, $r_{\max} = 400$ m;
- a large environmentally-driven downburst, characterized by $z_{\max} = 120$ m and $r_{\max} = 5500$ m.

Depending on the size of the downburst relative to the bridge, two extreme scenarios can be considered: (i) the downburst size is comparable to or larger than the bridge, or (ii) the downburst size is smaller than the bridge. In the first case, the three-dimensional field provided by the model, constituted by a vertical jet dragged by the environmental wind, invests all the bridge and simulates variations in wind speed in all directions. Specifically, apart from the bridge deck cross-section aligned with the jet's longitudinal axis, the other sections are impacted by an inclined wind velocity vector with both vertical and yaw angles of attack. When the downburst size is smaller than that of the bridge, in the second case, the deck sections external to the area invested by the downburst are considered immersed in a uniform flow characterized by the environmental wind velocity.

In several cases, a comparison with Oseguera-Bowles-Vicroy's model has been conducted [27–29]. The model recast with the current notation is reported in Appendix.

4.1. Flow patterns and velocity fields

Firstly, we show the flow pattern and analyse the velocity field of small-diameter downbursts, stationary or environmentally-driven, characterized by $z_{\max} = 25$ m, $r_{\max} = 400$ m, $\mathcal{U}_{\max} = 40$ m/s and $U_t = 13.3$ m/s.

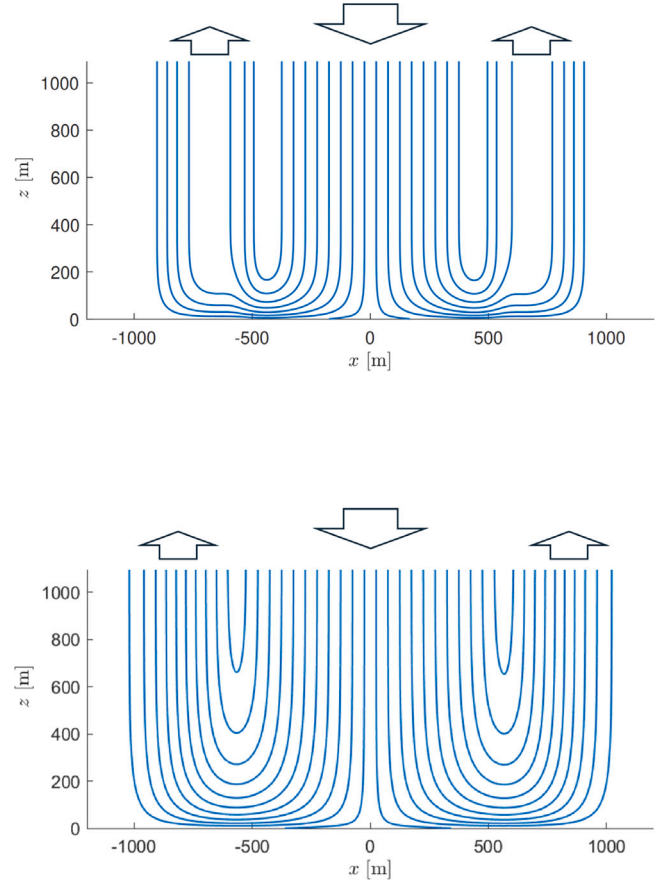


Fig. 4. Streamlines in case of a small-diameter stationary downburst (no environmental flow): proposed model (top); Vicroy's model (bottom). Input parameters: $z_{\max} = 25$ m, $r_{\max} = 400$ m and $\mathcal{U}_{\max} = 40$ m/s.

Fig. 4 shows the streamlines of the stationary downbursts, obtained through the present and Vicroy's model [29].

A tube of coherent vertical streams, like the nozzle of a jet flow, can be identified in both flow fields. A narrower tube is predicted by the proposed model which starts to expand and reverse flow direction at a lower height from the ground compared to that of Vicroy's model. Diameters of the jet of 440 m and 565 m can be, respectively, identified for the present and Vicroy's model. They correspond to ratios $r_{\max}/D$ of 0.91 and 1.41, respectively. It can be observed that the former value is in good agreement with the field observations [7] while the latter one is very close to the upper extreme of the interval recognized in the literature [36,39].

In addition, the streams near the ground show a sort of "bulge" at around 500 m from the centre of the tube in accordance with the behaviour shown in the plot at the top of Fig. 2. Similar behaviour is depicted in the well-known sketch deduced from field observation (Fig. 1).

Finally, in contrast to Vicroy's model, the streamlines of the present model near the ground show a smaller region with horizontal flow.

As expected, a more complex field can be observed by considering environmentally driven downbursts (Fig. 5). The smoother flow field of Vicroy's model is immediately apparent. In both fields, a significant inclined flow is recognizable instead of the tube characterizing the stationary downbursts. In addition, the streamlines of the present model show a clock-wise vortex after the passage of the jet and some dips on both sides of Fig. 5, both produced by the local increase in velocity depicted in the plot at the top of Fig. 2.

The three-dimensional velocity fields produced by small-diameter stationary and environmentally-driven downbursts are shown in Fig. 6.

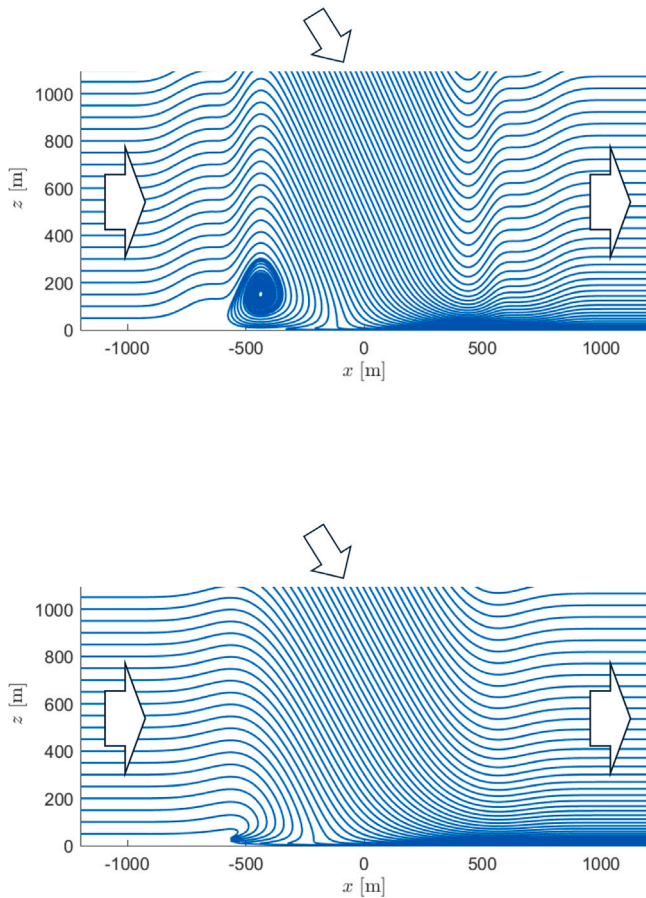


Fig. 5. Streamlines in case of a small environmentally-driven downburst: proposed model (top); Vicroy's model (bottom). Input parameters: $z_{\max} = 25$ m, $r_{\max} = 400$ m, $U_{\max} = 40$ m/s and $U_i = 13.3$ m/s.

Vertical and horizontal plane sections are used to show the vector fields. The stationary downburst (top of Fig. 6) appears as a jet flow directed towards the ground in which vertical vectors highlight the presence of the flow tube while the flow spreads radially (see Fig. 4 for comparison). The driven downburst (bottom of Fig. 6) shows inclined velocity vectors inside the tube and higher values of the flow velocity in the direction of the environmental wind. This behaviour is confirmed by the streamlines illustrated in Fig. 5.

The flow field is also analysed by considering the flow velocity and angle of attack during the passage of the small-diameter downburst. The time variation of the modulus of the flow velocity (red line) is superimposed to the angle of attack (blue line) in (Fig. 7) for a point at $z_{\text{deck}} = 60$ m over the ground. The results are plotted by considering time increasing from the right to the left, in agreement with the registration reported in Fig. 1. Starting from the right side of the plot, the modulus of the flow velocity corresponds to the environmental velocity (V_i). At around 80 s, the field point is reached by the downburst showing a rapid increase and sudden decrease of the velocity around the peak value, corresponding to about 48 m/s. A second smaller peak is observed followed by a strong reduction of the velocity reaching almost the status of quiet at the considered point. The last part of the signal shows a recovery of the velocity up to the environmental value (downburst passed). It is important to highlight the non-stationary behaviour of the flow field. The entire passage of the simulated downburst takes roughly 150 s with only 75 s for the development of the main peak.

Concerning the angle of attack, small fluctuations characterized by a negative angle of attack precede a rapid increase up to about 90°

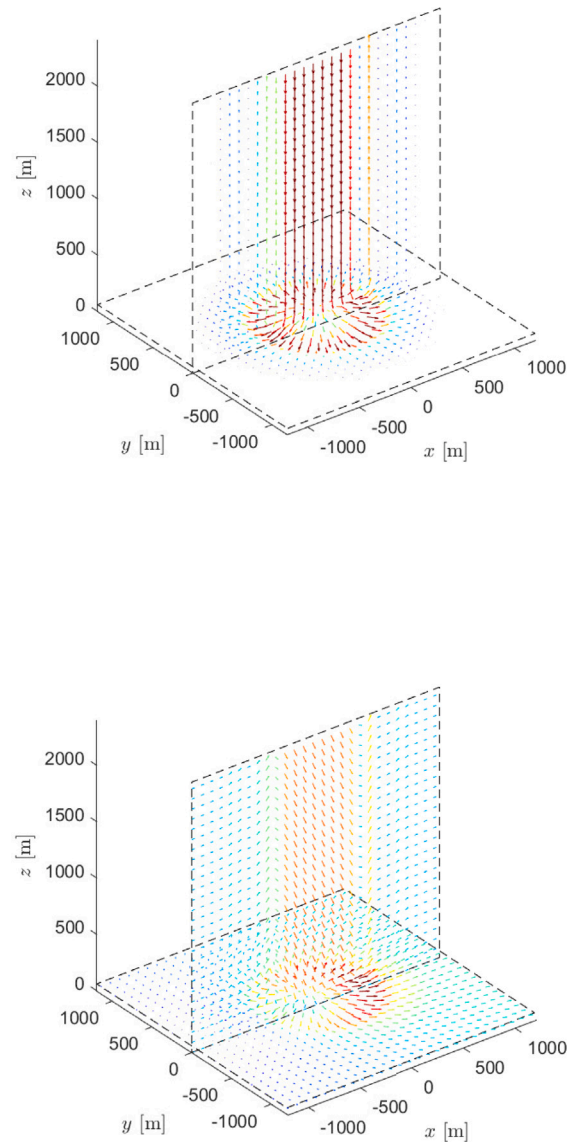


Fig. 6. Three-dimensional velocity field of the simulated downbursts: stationary (top) and environmentally-driven downbursts (bottom); input parameters: $z_{\max} = 25$ m, $r_{\max} = 400$ m, $U_{\max} = 40$ m/s and $U_i = 13.3$ m/s.

followed by an abrupt decrease of roughly 180° . The horizontal flow velocity is then reached after the development of another fluctuation. From the design point of view, it is apparent that the classical range of angles of attack measured in wind tunnel tests, from -10° to $+10^\circ$, is no longer suitable for characterizing the wind loads during the passage of the downburst. The practical implications of this requirement are relevant considering the significant increase in the blockage ratio of the wind tunnel test chamber obtained when a sectional model reaches angles of attack near 90° , especially for streamlined bridge decks. An additional sectional model with a very small scale is required for the common wind tunnel facilities.

Finally, the grey area in the plot highlights the time interval in which the angle of attack is over 10° , i.e. the upper extreme of the classical range, which corresponds to the high value of the modulus of the wind velocity. As will be shown later, in these time intervals the wind loads correspond to the maximum ones, not predictable with the data available in classical wind tunnel tests on sectional models. It is worth noting that the peak wind speed corresponds to an angle of attack of 10 degrees, which demonstrates that the wind loads during the

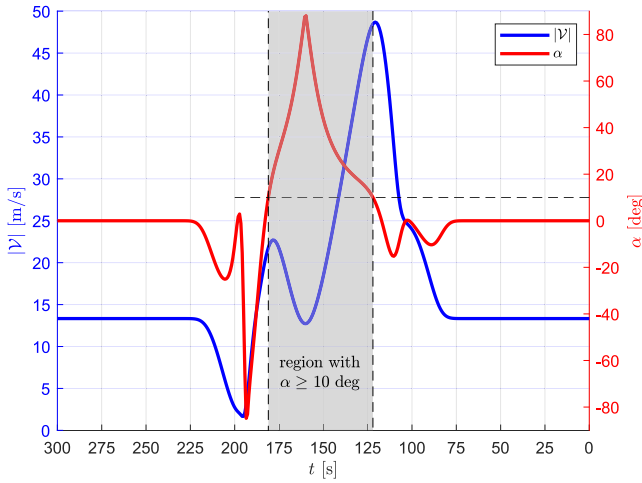


Fig. 7. Modulus of the wind velocity and angle of attack in case of a small environmentally-driven downburst with $z_{\max} = 25$ m, $r_{\max} = 400$ m, $U_{\max} = 40$ m/s, $U_i = 13.3$ m/s and $z_{deck} = 60$ m. Present model.

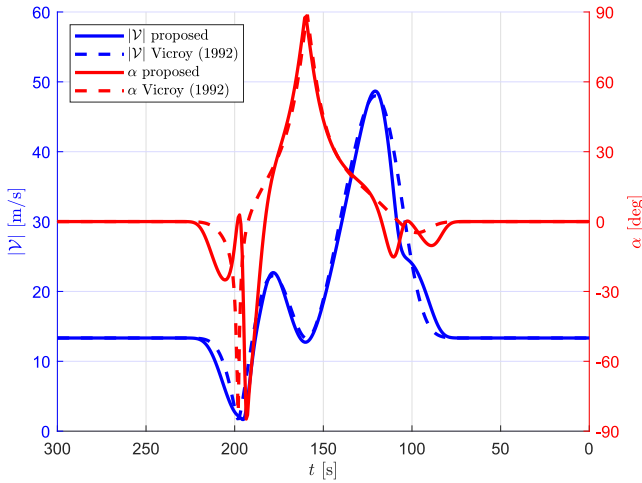


Fig. 8. Modulus of the wind velocity and angle of attack in case of a small environmentally-driven downburst with $z_{\max} = 25$ m, $r_{\max} = 400$ m, $U_{\max} = 40$ m/s, $U_i = 13.3$ m/s and $z_{deck} = 60$ m: proposed model (solid lines); Vicroy model (dashed lines).

passage of a downburst exceed the corresponding synoptic actions. In fact, it is well known that the aerodynamic coefficients of streamlined bridge decks significantly increase by changing the inclination of the flow.

The same plot is reported in Fig. 8 where the dashed lines refer to Vicroy's model. A good agreement between the two models is evident except for the time when the local increase in velocity depicted in the plot at the top of Fig. 2 reaches the bridge deck. A similar comparison is observed in terms of variation in the angles of attack predicted by the two models. Although significant differences in terms of maximum wind loads on the deck are not recognizable from the analysis of these behaviours, the evaluation of the extent to which the additional oscillations predicted by the present model influence the dynamic responses of the bridge deserves further investigation.

Fig. 9 shows the flow pattern predicted by the present model for a large environmentally-driven downburst characterized by $z_{\max} = 120$ m and $r_{\max} = 5500$ m, maintaining the same downburst ($U_{\max} = 40$ m/s) and environmental ($U_i = 13.3$ m/s) wind velocities. A much larger domain is used to capture the salient details of the phenomenon. Here the core of the vortex behind the tube is located at about 6000 m,

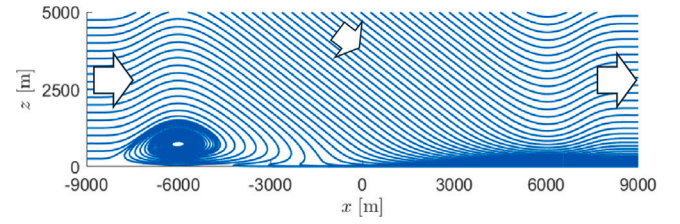


Fig. 9. Streamlines in case of a large-diameter environmentally-driven downburst. Input parameters: $z_{\max} = 120$ m, $r_{\max} = 5500$ m, $U_{\max} = 40$ m/s and $U_i = 13.3$ m/s. Present model.

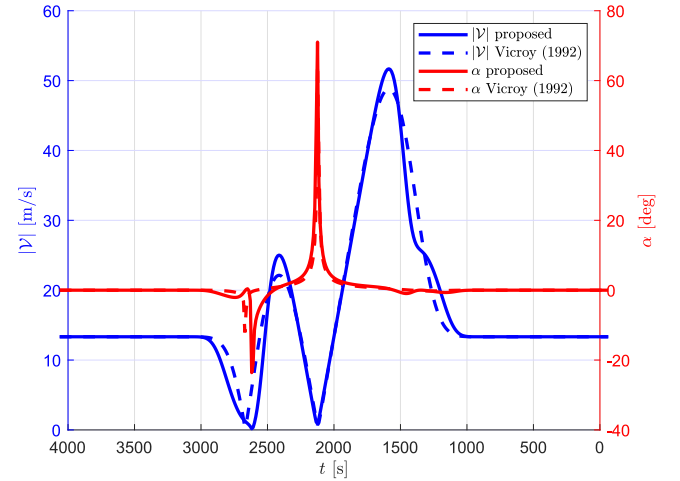


Fig. 10. Modulus of the wind velocity and angle of attack in case of a large environmentally-driven downburst with $z_{\max} = 120$ m, $r_{\max} = 5500$ m, $U_{\max} = 40$ m/s, $U_i = 13.3$ m/s and $z_{deck} = 60$ m: proposed model (solid lines); Vicroy model (dashed lines).

while for the small environmentally-driven downburst this distance is less than 500 m as shown in Fig. 5(a). Moreover, an inclined flow is again apparent without the anticipating and following fluctuations recognizable in the plot at the bottom of Fig. 5. Thus, lower fluctuating actions on the bridge are expected during the passage of a large downburst anticipating a different dynamic response.

The flow velocity and angle of attack during the passage of the large-diameter downburst is also analysed as done for the small-diameter environmentally-driven downburst. Fig. 10 shows the time variations of both quantities at a deck level of 60 m by using the present and the Vicroy model (see also Fig. 8 for comparison). It is underlined that the angle of attack peaks in a very limited time interval in correspondence with a flow velocity very close to zero. In this case, the load generation mechanism is no longer based on the combination of the wind velocity with the angle of attack as for a small downburst. In addition, the results of the two models are very similar except for the lower maximum peak provided by Vicroy's model. Finally, no fluctuations before and after the peak of the angle of attack are observed.

4.2. Aerodynamic forces

We adopt a simplified load model based on a sectional quasi-steady approach. A scheme of the aerodynamic behaviour of the deck is reported in Fig. 11 [40].

Unsteady effects and vortex shedding are not considered for the sake of simplicity. The short duration of the passage of the downburst over the bridge deck suggests the absence of the build-up time for

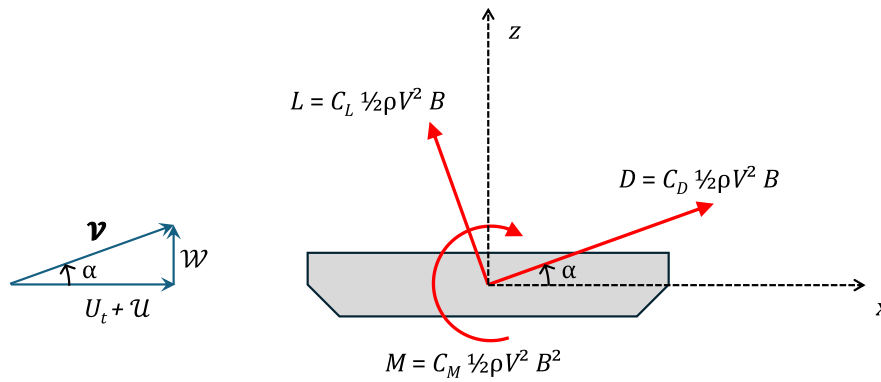


Fig. 11. Scheme of cross-sectional aerodynamics.

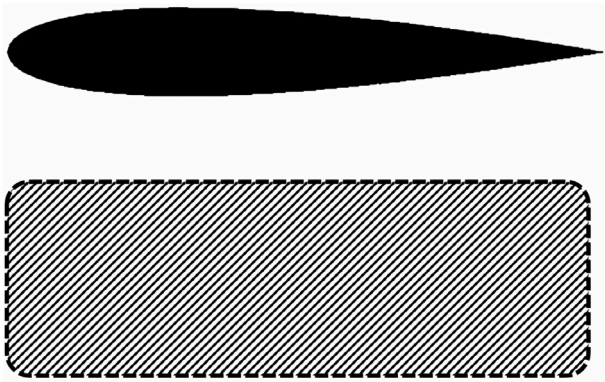


Fig. 12. Cross sections: NACA 0015 wing (top) and pseudo-rectangular 3:1 profile with rounded edges (bottom).

vortex induced vibrations [41,42]. Unsteady effects are of course possible and shall require sophisticated wind tunnel simulations for their assessment.

As highlighted above, even sticking with the simplified quasi-steady approach, the simulation of the aerodynamic forces on a bridge deck during the passage of a downburst requires the knowledge of the aerodynamic coefficients for wider ranges of angles of attack. Unfortunately, the current aerodynamic design approach, based on synoptic winds, considers only the classical range of $\pm 10^\circ$. As a consequence, no data are available in the literature for the aerodynamic coefficients of bridge decks for a very large range of angles of attacks. Therefore, it has been necessary to look for data in the literature on aircraft for which extensive wind tunnel tests have been conducted on wing profiles. Thus, the drag, lift and moment coefficients of the NACA 0015 wing reported in [43] are used for the first simulations (Fig. 12). Fig. 13 shows the aerodynamic coefficients for the entire range of angles of attack derived starting from the data corresponding to the 180-degree angle of attack range and accounting for the symmetry of the sections. It is evident the classical behaviour of a wing: the very low value of the drag coefficient and a significant increase of the lift coefficient for small variation around zero degrees.

In opposition to the lifting aerodynamic behaviour of the wing, a cross-section with a prevalent drag coefficient has been selected. In particular, a pseudo-rectangular cross-section with a width-to-depth ratio of 3 tested in [44] for the entire range of angles of attack has been used for a second set of simulations. The aerodynamic coefficients are reported in Fig. 14.

4.3. Simulation of the wind actions

The variations with time of the horizontal and vertical forces per unit length at the deck level ($z_{deck} = 60$ m) as well as the moment

normalized by the deck width ($B = 40$ m) are determined during the passage of the small and large-diameter downbursts.

The results for the deck with NACA 0015 cross section are reported in Fig. 15. It is immediately apparent that the vertical force corresponding to the small downburst is an order of magnitude higher than that of the large-diameter downburst. This effect can be explained by considering the results reported in Fig. 8 for the small-diameter downburst and in Fig. 10 for the large one. In the latter, large angles of attack are only observed for a very limited time interval characterized by a small wind velocity. Then, the aerodynamic forces during the passage of the large downburst are only produced by the drag coefficient, which is very low for a wing profile. The same motivation can be used to explain the prevalence of the vertical forces on the drag and moment observed for both downbursts. Interestingly, the small downburst provides the non-negligible vertical loads of 1.62 kN/m^2 . In addition, it can be also observed that the vertical force passes from the positive peak to the negative one in just 30 s, which demonstrates the non-stationary behaviour of the phenomenon. Conversely, a much longer time interval (about 500 s) is required for the large downburst. Finally, the small and large downbursts last approximately 160 s and 2000 s, respectively. Taking into account the environmental velocity of 13.3 m/s , this translates to horizontal extensions of about 2.2 km and 26 km for the small and large downbursts, respectively.

Fig. 16 shows the time variation of the forces obtained by considering the pseudo-rectangular 3:1 cross-section. Some differences are recognizable by employing this bluff cross-section. First, for the small-diameter downburst, the horizontal force becomes comparable to the vertical one due to the significant drag coefficient of the rectangle (top plot). In contrast, the horizontal force prevails in the plot of the large-diameter downburst (bottom plot) because significant angles of attack are absent during the occurrence of high flow velocities. It is worth noting that, the vertical force reaches a significant peak value (over 1 kN/m^2) also for the pseudo-rectangular cross-section.

The time histories of the forces reported in Figs. 15 and 16 provide a partial picture of the wind loads on bridge decks invested by downbursts, as they refer to a single deck elevation. For the same downburst, the strong three-dimensional behaviour of the phenomenon requires time-force plots for several values of the deck height. In addition, the uncertainties in the definition of the design downburst for a specific site make it necessary to compare the results of different assumptions on its parameters.

Therefore, simulations of the aerodynamic forces have been conducted for several values of r_{max} and z_{max} registering only the peak forces along the height. In particular, Fig. 17 shows the variation along the height of the maximum absolute values of the force coefficients $C_F = F / (\frac{1}{2} \rho U_{max}^2 B)$ with $F = F_x, F_z$, and $C_M = M / (\frac{1}{2} \rho U_{max}^2 B^2)$ for the NACA profile. Several downburst dimensions are considered by varying r_{max} (along rows and columns of diagrams) and z_{max} (along rows of diagrams).

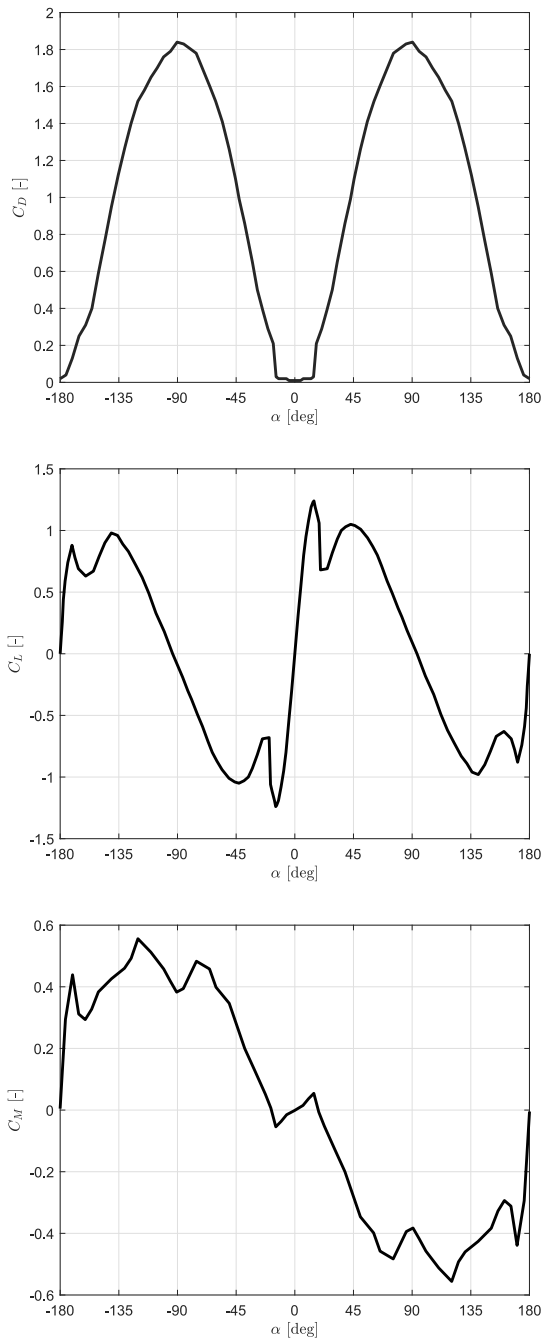


Fig. 13. Drag (C_D), lift (C_L) and moment (C_M) coefficients of the NACA 0015 wing for the entire range of angles of attack [43].

As observed for Fig. 15, the vertical force coefficients prevail both on the horizontal and moment coefficients. The maximum vertical force is obtained for a downburst of intermediate radius ($r_{\max} = 2000$ m, with C_F of about 1.75) for a deck located at around 150 m over the ground. At 60 m, the smallest downburst provides a very similar action characterized by a C_F of about 1.65. As expected, for the larger downbursts, the horizontal and moment coefficients almost disappear from the plot.

The same analysis is repeated for the pseudo-rectangular section and its results are reported in Fig. 18. In the case of small- and medium-diameter downbursts, noteworthy horizontal force coefficient profiles intersect with vertical force coefficient profiles at specific heights, varying depending on the particular downburst under consideration.

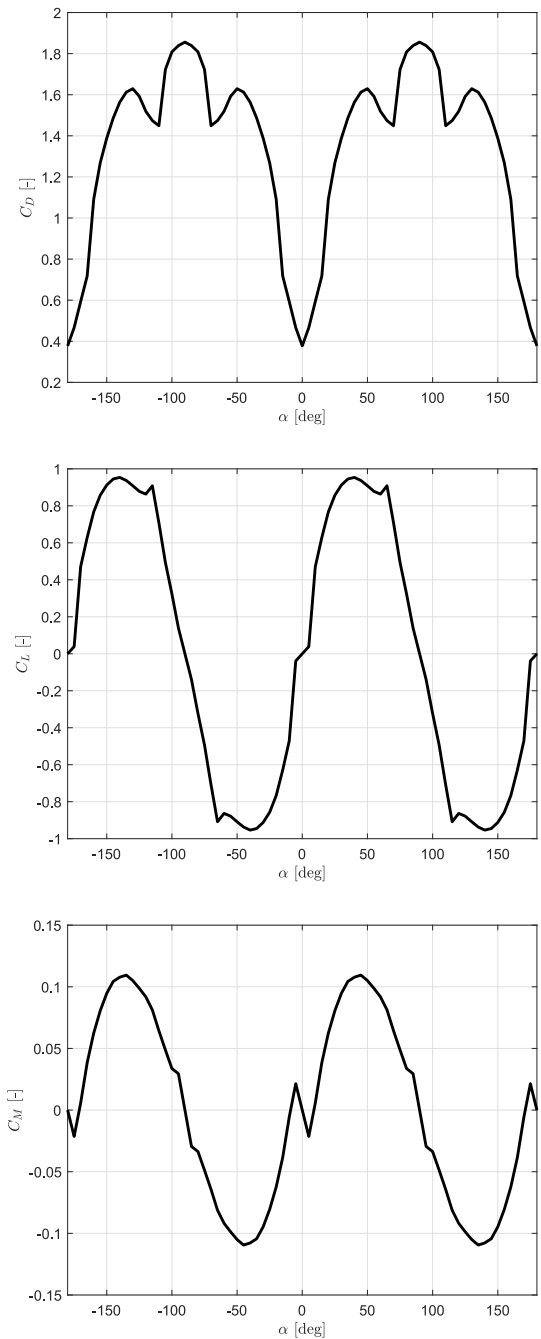


Fig. 14. Drag (C_D), lift (C_L) and moment (C_M) coefficients of a 3:1 pseudo-rectangular cylinder with rounded edges for the entire range of angles of attack. Source: Adapted from [44].

Conversely, the horizontal force coefficient profile of large downbursts prevails at all heights.

In the absence of site-specific studies, the plots in Fig. 17 or 18, developed for the deck section under design, can serve as a valuable tool for incorporating downburst actions into the bridge deck assessment. Fixing the deck height of the specific bridge, it provides the maximum force coefficients along several downburst dimensions.

5. Conclusions

The proposed three-dimensional model improves upon existing models by incorporating the observed local increase in velocity from

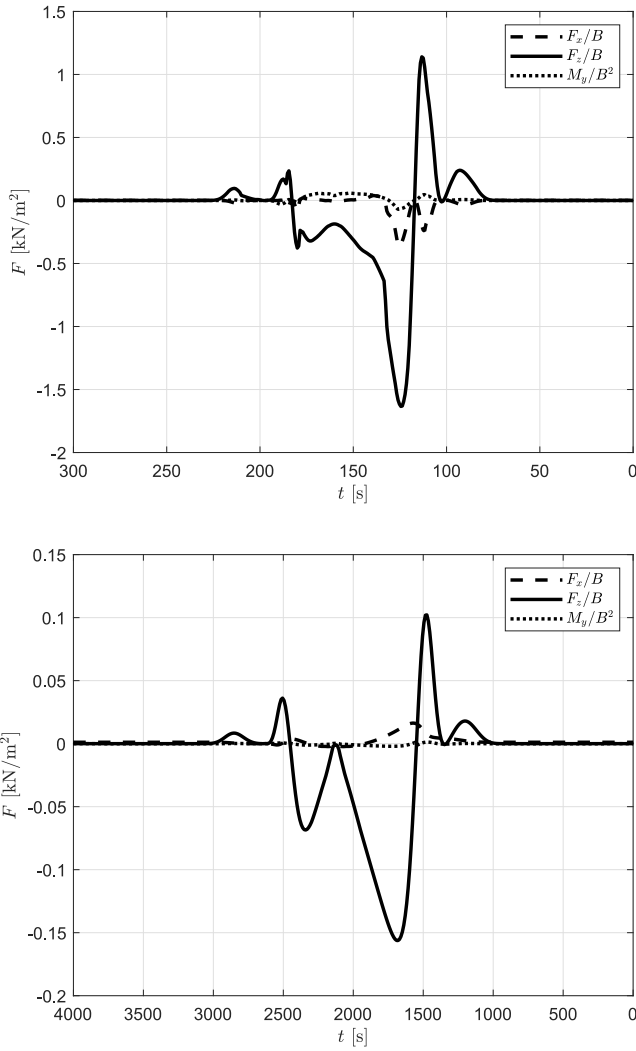


Fig. 15. Horizontal and vertical forces per unit length on a 40-m-wide deck with NACA 0015 cross-section: small (top) and large-diameter downburst (bottom). Input parameters: $z_{\max} = 25$ m and $r_{\max} = 400$ m (small downburst), $z_{\max} = 120$ m and $r_{\max} = 5500$ m (large downburst), $U_{\max} = 40$ m/s, $U_i = 13.3$ m/s and $z_{\text{deck}} = 60$ m.

average field measurements through a novel definition of the radial shaping function of radial velocity. Furthermore, the coefficients of an existing empirical vertical shaping function of horizontal velocity have been recalibrated to better match available field measurements

The analytical model is first used to depict the flow patterns for stationary and environmental-driven downbursts characterized by a small diameter. In the stationary case, the streamlines show the expected pattern of a stationary jet in which the effects of the introduction of the “bulge” in the shaping functions are also recognizable. The dimension of the tube is more in line with the field measures than the literature models. In addition, the present model shows a reduced area characterized by only horizontal flows compared to Vicroy’s model.

When the downburst is driven by the environmental wind, flow regions characterized by significant inclinations are evident, leading to large variations in the angle of attack of the flow approaching the deck. The present model also shows a vortex after the passage of the inclined jet along with some dips on both sides, whose impact on the dynamic response of flexible bridges should be verified.

The effect of the terrain in shaping the environmental-wind-driven downburst has been neglected and could represent an interesting addition to the model. However, when considering bridges over the sea, the

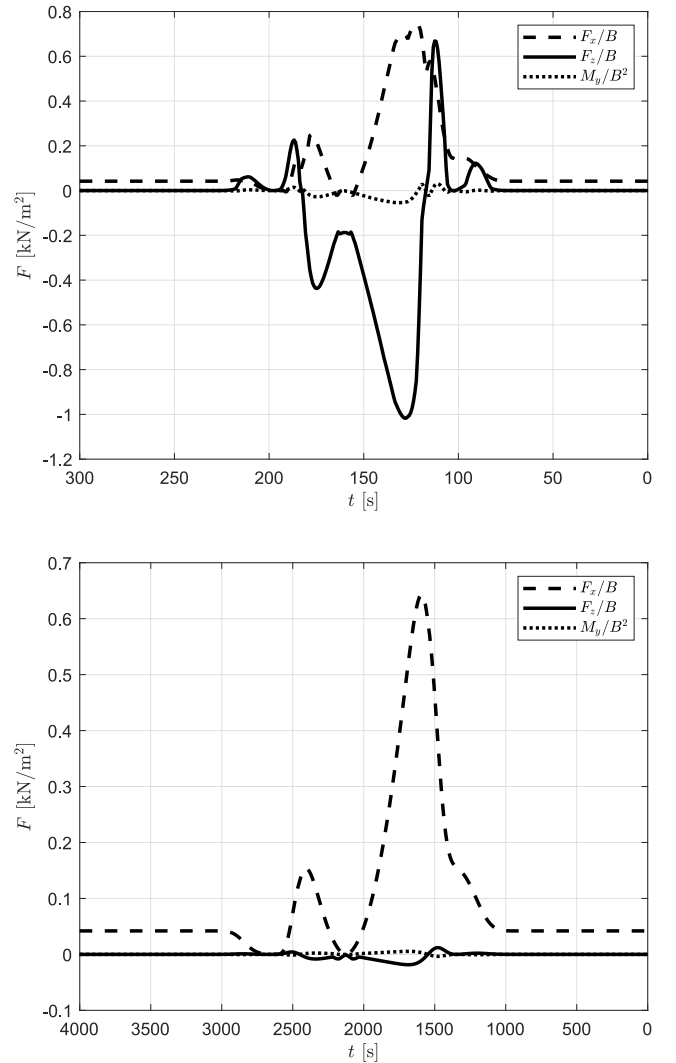


Fig. 16. Horizontal and vertical forces per unit length on a 40-m-wide deck with pseudo-rectangular 3:1 cross-section: small (top) and large-diameter downburst (bottom). Input parameters: $z_{\max} = 25$ m and $r_{\max} = 400$ m (small downburst), $z_{\max} = 120$ m and $r_{\max} = 5500$ m (large downburst), $U_{\max} = 40$ m/s, $U_i = 13.3$ m/s and $z_{\text{deck}} = 60$ m.

effect of terrain roughness may be negligible at the level of the bridge deck.

Investigations of the wind field at different deck levels for small to medium-diameter downbursts reveal the presence of non-negligible time intervals during which the angle of attack is over 10 degrees with high wind velocities. Force simulations confirm that the wind loads reach their maximum during these intervals, which are not predictable with existing approaches. Given that in most regions the wind velocity associated with downbursts is of the same magnitude as the one of synoptic winds, the results demonstrate that the wind loads during the passage of a downburst exceed the corresponding synoptic actions, especially for streamlined bridge decks in which the aerodynamic coefficients significantly increase by changing the inclination of the flow.

In conclusion, the results of this study underscore the importance of including the effects of downbursts in the aerodynamic design of bridges. It is also highlighted that the current range of angles of attack investigated in wind tunnel tests for estimating the aerodynamic coefficients should be extended to supply the input data for downburst simulations. For conventional wind tunnel facilities, this requirement

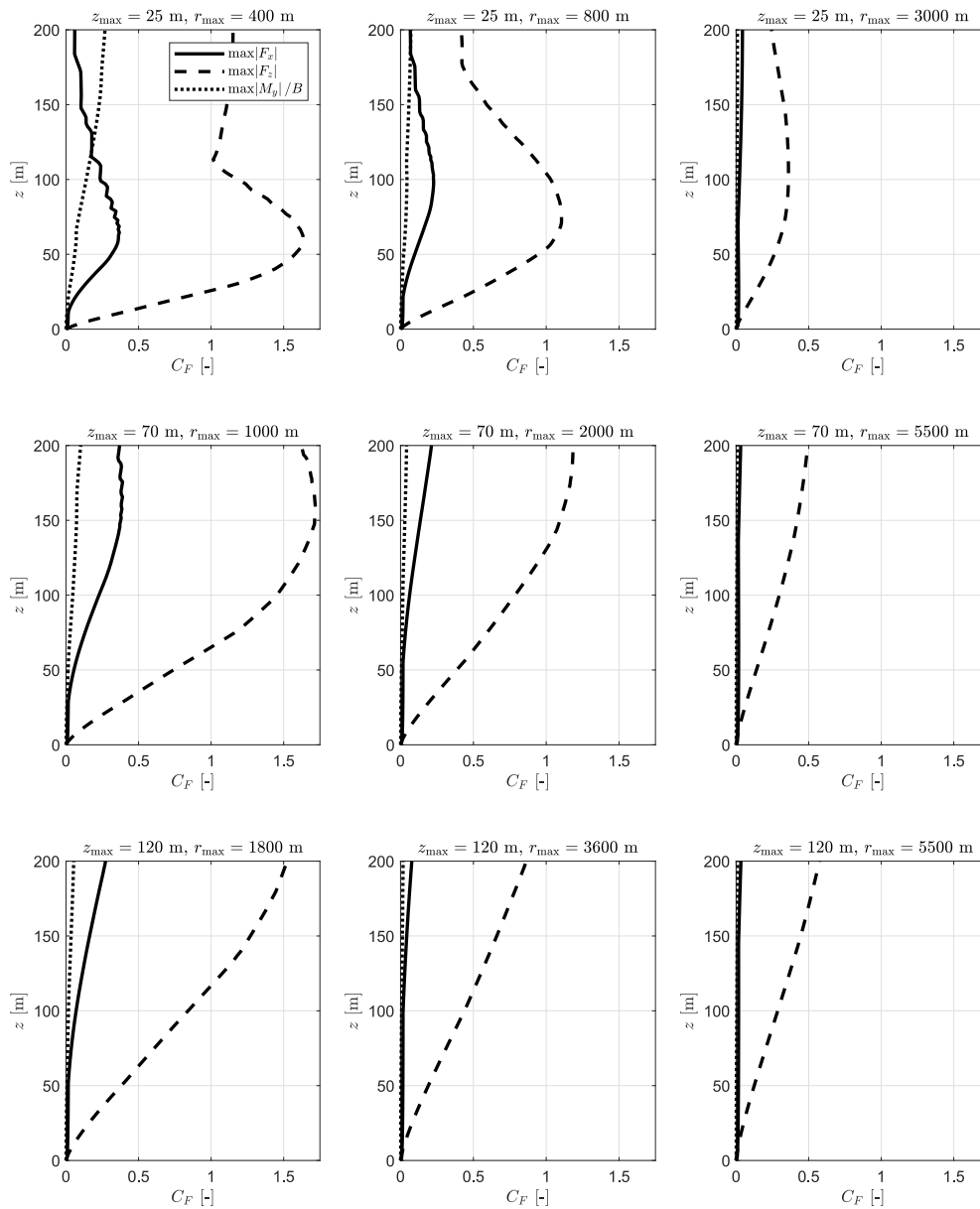


Fig. 17. Peak forces varying the deck height for the NACA 0015 profile and different downbursts.

necessitates the use of a highly scaled sectional model. This is particularly crucial given the noticeable increase in the blockage ratio at large angles of attack, especially when studying streamlined bridge decks.

The quantitative results in terms of forces are, of course, influenced by the lack of aerodynamic characterizations of bridge cross-sections at large angles of attack, so that here only two limit cases have been presented.

CRediT authorship contribution statement

Luca Salvatori: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Antonino Maria Marra:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mario De Stefano:** Supervision, Funding acquisition. **Paolo Spinelli:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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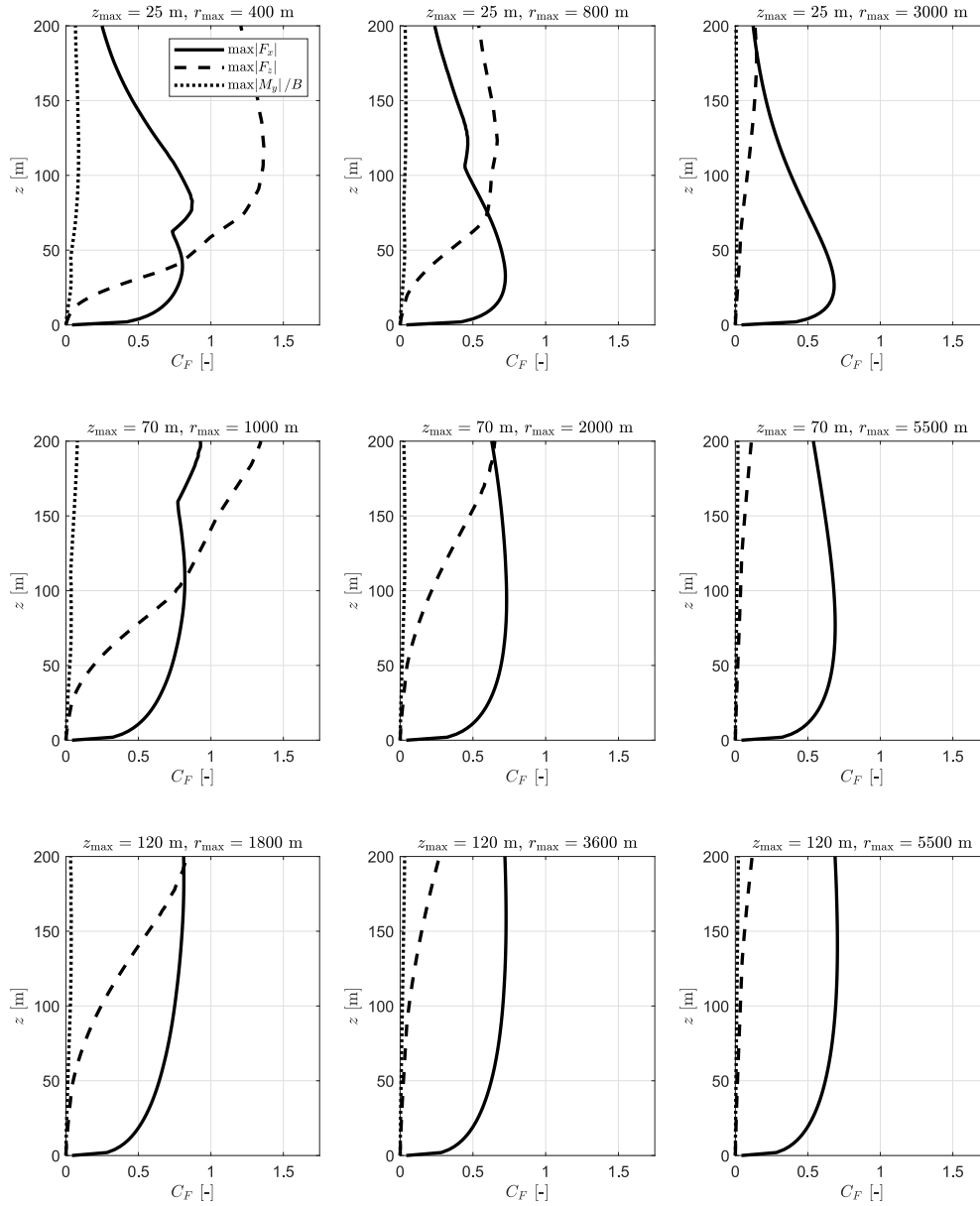


Fig. 18. Peak forces varying the deck height for the pseudo-rectangular 3:1 cross-section and different downbursts.

Appendix. Oseguera-Bowles-Vicroy's model

For the sake of comparison, we report the Oseguera-Bowles-Vicroy model [27–29] recast in the form we adopted.

The radial modulation of the horizontal velocity is written as

$$v_\rho(\rho) = \frac{1}{\hat{\omega}_\rho} \frac{\rho}{2} e^{\frac{2-\rho^{2\alpha}}{2\alpha}}$$

where $\hat{\omega}_\rho = \frac{1}{2} e^{\frac{1}{2\alpha}}$ is a normalization coefficient and α a shape parameter. The value of $\alpha \approx 2$ is suggested in [28], from which $\hat{\omega}_\rho \approx 0.6420$.

The corresponding radial modulation of the vertical velocity is

$$\omega_\rho(\rho) = \frac{1}{\hat{\omega}_\rho} \left(1 - \frac{1}{2} \rho^{2\alpha} \right) e^{\frac{2-\rho^{2\alpha}}{2\alpha}}$$

with $\hat{\omega}_\rho = e^{\frac{1}{\alpha}} \approx 1.6487$.

As to the vertical modulation of the horizontal velocity the form

$$v_\zeta(\zeta) = \frac{1}{\hat{\omega}_\zeta} (e^{c_1 \zeta} - e^{c_2 \zeta})$$

is suggested. In order the maximum to be at $\zeta = 1$, one must have $\ln \frac{c_1}{c_2} = c_2 - c_1$, and the normalization coefficient is $\hat{\omega}_\zeta = \left(\frac{c_1}{c_2} \right)^{-\frac{c_1}{c_1 - c_2}} - \left(\frac{c_1}{c_2} \right)^{-\frac{c_2}{c_1 - c_2}}$. The authors suggest $c_1 \approx -0.15$, $c_2 \approx -3.2175$, from which $\hat{\omega}_\zeta \approx 0.8207$.

The corresponding vertical modulation of the vertical components becomes

$$\omega_\zeta(\zeta) = \frac{1}{\hat{\omega}_\zeta} \left(\frac{e^{c_1 \zeta} - 1}{c_1} - \frac{e^{c_2 \zeta} - 1}{c_2} \right)$$

with $\hat{\omega}_\zeta = \frac{1}{c_2} - \frac{1}{c_1} \approx 6.3559$.

Finally, the relationship between the maximum horizontal velocity and the downdraft velocity is

$$\mathcal{W}_{\max} = \frac{\hat{\omega}_\rho \hat{\omega}_\zeta}{\hat{\omega}_\rho \hat{\omega}_\zeta} \frac{z_{\max}}{r_{\max}} \mathcal{U}_{\max} \approx 19.89 \frac{z_{\max}}{r_{\max}} \mathcal{U}_{\max} \quad (\text{A.1})$$

It can be observed that the downdraft velocity is almost twice as large as the one in the model developed in the present paper (compare Eqs. (18) and (A.1)); however, the final fields do not differ significantly

at the heights of interests because the different values of the modulation function ω_c (see Fig. 3).

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